



## Jialing Dai

Ph.D. Candidate in Computer Science | Ph.D. expected Oct. 2026  
University of Chinese Academy of Sciences | Visiting Scholar, UT Dallas  
Problem-Driven Modeling | Provable Optimization | Sampling/Coverage Algorithms | GNN & Deep RL  
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- **Research profile: problem-driven computational modeling and optimization.** I start from problem discovery, identify computable structure in real phenomena, and turn it into formal models, provable objectives, scalable algorithms, and empirically testable claims.
- **Publication record:** 6 published/accepted peer-reviewed papers, including two first-author Information Sciences papers, one Chinese Journal of Management Science paper, two AAIM 2026 papers, and one ICRA 2025 co-authored paper; first-author IEEE/ACM Transactions on Networking major revision submitted; ESWA major revision in progress.
- **Method base: mathematical modeling, NP-hardness and approximation analysis, non-submodular/non-monotone objectives, sampling/greedy/coverage algorithms, GNN/deep-RL implementation, and reproducible experiments.**
- **International collaboration: visiting researcher at the University of Texas at Dallas from Jan. 2025 to Jan. 2026, working with Prof. Weili Wu and Prof. Ding-Zhu Du on algorithmic modeling, graph algorithms, robust coverage/provenance, and data-driven validation.**
- **Career fit: postdoctoral, research fellow, and research-oriented data/AI roles requiring formal modeling, optimization, learning-based solution methods, and empirical validation across decision systems.**

### Education

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| 2020.09 – Present | <b>University of Chinese Academy of Sciences</b> <b>Ph.D. in Computer Science (Computer Application Technology)</b><br>Advisor: <a href="#">Prof. Jianming Zhu</a> ; expected defense in mid-Aug. 2026; degree expected in late Sept./early Oct. 2026.<br><b>Dissertation:</b> Formal Modeling and Optimization Algorithms for Group Opinion Dynamics and Influence-Related Decision Problems; GPA: 3.83/4.00 |
| 2025.01 – 2026.01 | <b>University of Texas at Dallas</b> <b>Visiting Researcher</b><br>Advisors: <a href="#">Prof. Weili Wu</a> and <a href="#">Prof. Ding-Zhu Du</a>                                                                                                                                                                                                                                                             |
| 2016.09 – 2020.06 | <b>Beijing University of Chemical Technology</b> <b>B.Mgmt. in Information Management and Information Systems</b><br>GPA: 92.68/100; ranked 1st in major; interdisciplinary training in information systems, data analytics, operations research, and management.                                                                                                                                             |

### Selected Research Work

- [1] **Jialing Dai**, Jianming Zhu, Guoqing Wang.  
"Competitive net influence maximization on intergroup debate effect."  
**Information Sciences**, 2024, vol. 680, p. 121139. **Focus:** hypergraph modeling, sampling algorithm design, and data-driven approximation. Formalized intergroup debate by coupling group-consensus nodes and member nodes in a hypergraph; defined the IDIC model and CNIM problem; proved NP-hardness and non-submodularity. Extended the setting to HCIC, used Sandwich-style upper/lower bounds without claiming to originate the framework, designed heterogeneous competitive influence sampling with greedy selection, and evaluated scalability on simulated and real datasets.
- [2] **Jialing Dai**, Jianming Zhu, Guoqing Wang.  
"Opinion influence maximization problem in online social networks based on group polarization effect."  
**Information Sciences**, 2022, vol. 609, pp. 195-214. **Focus:** physics-inspired preference dynamics modeling and empirical validation. Built a real OSN dataset (50 posts; 127,178 replies) to identify group polarization from data; constructed the EIC model from Ising-model-inspired preference updating and designed GEIM to model the joint effect of persuasive exposure and preference divergence; experiments analyze how initial preferences, group size, and opposing opinions affect seed selection.

- [3] **Jialing Dai**, Yisheng Zhou, Yefeng Sun, Jianming Zhu, Weili Wu.  
"Unified Group-Aware Influence Maximization with Generalized Deep Reinforcement Learning."  
**IEEE/ACM Transactions on Networking**, major revision submitted. **Focus:** unified mechanism abstraction and GNN/deep-RL optimization. Built UGIP for echo chamber, topic-aware, and related group-aware diffusion mechanisms; analyzed non-submodularity/non-supermodularity and inapproximability; designed UG-DCGNN with coupled GNN, DDQN, and group features to reduce inference-time Monte Carlo simulation.
- [4] **Jialing Dai**, Jianming Zhu, Guoqing Wang, Jun Huang.  
"Net positive information diffusion activity maximizing in signed online social networks."  
**Chinese Journal of Management Science**, 2025, 33(3): 139-150. **Focus:** objective reformulation and path-level coverage optimization. Shifted signed diffusion from node coverage to interaction-level positive/negative activity; proved NP-hardness, #P-hardness, and non-submodularity; constructed a path-level monotone submodular surrogate solved by maximum-coverage greedy optimization with CELF.
- [5] Ke Su, **Jialing Dai\*** (corresponding author), Weili Wu.  
"Scenario-Robust Checkpoint Cover for AI-Generated Content Provenance."  
**AAIM 2026, accepted; Springer LNCS**. **Focus:** robust coverage modeling for AIGC provenance. Formulated scenario-robust checkpoint cover, turning provenance tracing under multiple scenarios into robust coverage optimization; corresponding author.
- [6] **Jialing Dai**, Ke Su, Jianming Zhu, Weili Wu, Ding-Zhu Du, Peikun Ni.  
"Influence Maximization in Online Social Networks: Models, Algorithms, and Trustworthy Decision-Making."  
**Expert Systems with Applications**, major revision in progress. **Focus:** model taxonomy and trustworthy decision framing. Organized models, objectives, algorithms, validation needs, reliability issues, and future research directions.

## Additional Publications

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- [7] **Jialing Dai**, Jianming Zhu, Ding-Zhu Du. "Approximation Algorithms for Orthant Submodular Optimization." Working paper on approximation algorithms for orthant submodular optimization.
- [8] Yefeng Sun, **Jialing Dai**, Liang Gong et al. "CaMT: Class-aware Multi-level Training Distillation for Lightweight Semantic Segmentation in Apple Orchard." Artificial Intelligence in Agriculture, **major revision**.
- [9] Yefeng Sun, Liang Gong, **Jialing Dai** et al. "A Robust Semantic-Enhanced Framework for Multi-Robot SLAM Merging in Orchard." Journal of Field Robotics, **major revision**.
- [10] Yisheng Zhou, **Jialing Dai**, Dongyu Mao, Ke Su, Weili Wu. "Modeling Organizational Performance through Competitive Resource Diffusion." **AAIM 2026, accepted; Springer LNCS**.
- [11] Yefeng Sun, Liang Gong, **Jialing Dai**, et al. "S!BEV: Lightweight, Robust, and Precise SLAM-Oriented Segmentation Bird Eye's View Mapping Approach." ICRA 2025.
- [12] Yefeng Sun, Wei Zhang, **Jialing Dai**, et al. "Future Agriculture: Multi-Robot Collaborative Cloud-based Control and Scheduling Framework in Greenhouse." IROS 2026, **under review**.

## Teaching, Projects, and Service

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- **Academic service:** Program Committee Member, AAIM 2026; reviewer for IEEE/ACM ToN, IEEE T-CSS, and World Wide Web.
- **Teaching assistant, UCAS (2021-2023):** tutorials, grading, office hours, and project guidance for Mathematical Methods, Accounting and Cost Management, and Human Factors Engineering.
- **Project lead, State Grid Henan EPRI (2022):** led a 6-member team to build a 1,498-case labeled dataset and deliver association/clustering analysis with management recommendations.
- **Research and academic operations:** fault-tree modeling project, seminar organization, graduate defense/admission coordination, textbook expansion, and operations-research translation support.

## Awards and Honors

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- **Awards:** CAST Young Talent Support Project, Doctoral Student Program (first cohort, 2024); Graduate National Scholarship (2022); Undergraduate National Scholarship (2018, 2019); Outstanding Graduate of Beijing (2020); ISEM 2021 Outstanding Young Scholar Paper, Second Prize.

## Courses and Skills

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- **Courses and skills:** Data Structures & Algorithms; Optimization/OR; Machine Learning; Computational Modeling; Python/PyTorch/NetworkX; complexity/approximation; sampling/greedy/coverage; GNN/deep-RL; reproducible experiments.